1. **Course number and name**
   CHE 330. Polymer Science

2. **Credits, contact hours and categorization**
   Credit hours: 3; Contact hours: 3 lecture hours per week; Engineering

3. **Instructor's or course coordinator's name**
   Roshan Nemade, Ezinne Achinivu

4. **Textbook, title, author, publisher, year**
   Polymer Chemistry, Second Edition / Edition 2 by Paul C. Hiemenz, Timothy P. Lodge, Tim Lodge

5. **Specific course information**
   a. Brief description of the content of the course (catalog description)
   Overview of polymer science and engineering. Introduction to polymers as an engineering material; basic concepts of properties; synthesis and structure of polymeric materials, polymer production, property prediction and performance in products.
   b. Prerequisites or co-requisites
      CHE 230 or CME 260 and CHEM 232.
   c. Indicate whether a required, elective, or selected elective.
      Required.

6. **Specific goals for the course**
   a. Specific outcomes of instruction.
   The main objective of the course is to familiarize the students with the fundamental concepts of Polymer Science from the lens of Material Science and Engineering which will be used as background knowledge for the understanding of specialized problems in the field of Chemical Engineering. Students will be able to:
      1. Understand the fundamental and essential concepts used in polymeric materials for Chemical Engineering applications.
      2. Know the detailed methods to synthesize and characterize the polymeric materials.
      3. Understand the properties that characterize the behavior of polymeric materials.
      4. Select appropriate types of polymeric material for specific application.
      5. Offer different approaches to modify structure/microstructure of polymeric materials to get desired properties

   One semester long project to reflect the three components of molecular engineering related to Polymer Design, Processing and Application. Students will be working with groups and are expected to prepare a presentation. Each group will be assigned one polymer to perform a critical review on the synthesis mechanisms followed by selecting a consumer product to explore products material and performance requirements.

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.			X	
(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.		X		
(3) an ability to communicate effectively with a range of audiences.		X		
(4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.		X		
(5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.		X		
(6) an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	X			
(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.		X		

7. Brief list of topics to be covered

- Introduction to Polymer Science
- Polymer Classification
- Methods and Intro to Mechanisms of Polymerization
- Characterization of polymers
- Thermodynamics of polymers
- Overview of Advanced characterization tools
- Polymer processing
- Polymer application